

Department of Electrical Engineering,
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Special Topic on Implementation
Research Report

FamilyPart: Training-free Open-
vocabulary 3D Hierarchical Part
Instance Segmentation via a Multi-level
Family Tree

以多層家族樹結構建立物件部件關係
之無需訓練開放語義三維實例分割

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Abstract

This training-free work proposes a novel approach to open-vocabulary 3D part-level instance segmentation. We first construct a family-like hierarchical structure by progressively segmenting images at multiple levels of granularity. Building on this structure, we design algorithms to retrieve 3D part-level instances after aggregating 2D masklets into 3D proposals, where cross-frame consistency is enforced by class-agnostic mask supervision between deep learning models.

Although the resulting mean average precision is not fully satisfactory, we identify the key failure modes and propose several directions for improvement. Overall, this work demonstrates the potential of developing more advanced pipelines using only posed RGB-D image sequences and 3D point clouds. Our method not only enhances hierarchical instance relationship understanding but also opens up new possibilities for embodied AI and robotic systems to perceive and interact with human-centric real-world environments.

摘要

本研究提出了一種無需額外訓練的三維開放語義部件層級實例分割方法。我們透過多個分割層級對影像進行漸進式細化，並利用深度學習模型之間的相互監督來建立幀間關聯性，隨後將二維遮罩串聚合為三維掩碼。最後，在所構建的家族樹狀階層關係基礎上，我們設計多種演算法以進行部件層級實例檢索。儘管在類別平均精度上的表現尚不理想，我們對失敗案例進行分析並歸納其成因，進而提出可能的改進方向。總而言之，本研究展現了在僅依賴彩色深度影像序列、相機姿態以及三維點雲的前提下，發展更先進處理流程的潛力。我們提出的方法不僅提升了對多階層實例關係的理解，也為實體人工智慧與機器人系統在感知並與以人類為中心的真實世界情境互動方面開啟了新的可能。

Table of Contents

1. Introduction.....	1
2. Related Works	2
3. Method	3
3.1 Multi-level Class-agnostic Family Tree Construction with Semantic-SAM.....	3
3.1.1 Solo Photo Method	4
3.1.2 Gap Filling	4
3.2 Provenance-aware SAM2 Tracking and Deduplication.....	5
3.2.1 Deduplicating and Virtual Children	6
3.2.2 Window-Based Method	7
3.3 3D Proposal Aggregation and Feature Extraction	8
3.3.1 Family-aware Proposal Merging.....	8
3.3.2 3D Proposal Feature Extraction	8
3.4 Open-Vocabulary Query Inference with Template	9
3.4.1 Hierarchical Strategy	10
3.4.2 Exhaustive Pairing Strategy	10
3.4.3 Independent Strategy	10
3.5 Evaluation with Search3D Metric.....	11
4. Experiments	11
4.1 Implementation Details.....	11
4.2 Results and Analysis	12
4.3 Future Directions	14
5. Conclusion	14
6. References.....	14
7. Project Management and Teamwork.....	16
7.1 Plan Management.....	16

7.2 Teamwork.....	16
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List of Illustrations

Fig. 1 Overview of the FamilyPart Pipeline.	3
Fig. 2 A family of instance in a single frame. The images from left to right are original frame, one L2 instance, seven L4 instances, five L6 instances.	4
Fig. 3 Comparison before and after gap filling; the rightmost image is applied gap filling.	4
Fig. 4 An example frame showing Semantic-SAM segments (middle) and the corresponding results after SAM2 tracking (right) in a single level.	5
Fig. 5 The diagram of Window-Based Tracking in a single level.	7
Fig. 6 An example of change in granularity after SAM2 tracking (at the chair). .	13

List of Tables

Table. 1 AP Results With Different Family Levels (laion-clip, min. area =500, prop=30) Unit: %	12
Table. 2 AP Results With Different Feature Extractor (L1/3/5, min. area=100) Unit: %	12
Table. 3 AP Results With Different Inference Strategies over 7 Experiments Conducted Unit: %	12

1. Introduction

In the future, the capability of robots and embodied AI to understand the semantic structure of 3D environments will be critical due to emerging applications and the need for intelligent robots to operate in human-centric spaces. Instance segmentation is one of the most significant tasks for robots to handle. To correctly interact with real-world objects, computer vision algorithms must obtain the exact location of an instance. In more complex scenarios, it is crucial to identify specific parts of an object. However, conventional 3D part-level instance segmentation pipelines are often constrained to a closed set of objects or parts, causing them to lose functionality outside of their training datasets.

Open-vocabulary 3D instance segmentation has emerged as a solution to this problem, as it enables the use of text queries to perform segmentation in arbitrary scenes. This eliminates the need to train or fine-tune a new model for novel environments. Furthermore, leveraging the flexibility of feature representations, it surpasses object-level constraints, gaining the capability to tackle a wide range of tasks at a more fine-grained part level.

The main novelty of this training-free work lies in several aspects. First, we propose a new method that constructs a hierarchical, family-like structure based on different granularities. Moreover, this structure can be refined and merged based on family relationships; therefore, it is beneficial for cross-level object and part retrieval. Second, this work aims to develop a new pipeline that does not require pre-segmented superpoint structures or 3D masks; that is, the entire pipeline depends solely on posed RGB-D image sequences and the 3D point cloud structure. Third, we aim to resolve the ambiguity inherent in coarse-to-fine segmentation backbone models, such as Semantic-SAM [9], used in this work. This goal is achieved by applying different cross-level strategies during instance retrieval.

The methodology developed in Section 3 represents my original contribution, formulated after discussing potential approaches with my mentors at the Vision Science Lab, Chen Chia-Min, Wang Yan-Qing, Yen Hong-Xuan and Ou Yeh. It is noted that Yen Hong-Xuan is the author of the related work PS3 [5], which is currently under review by WACV. Regarding the implementation details, the procedures in Section 3.1 for invoking the Semantic-SAM [9] API utilized code provided by Yen Hong-Xuan. Furthermore, for the 3D aggregation and SigLIP [11] feature extraction in Section 3.3, I adopted the pooling methods to PS3 [5] and OpenMask3D [2], while the corresponding code implementation was completed with assistance from Ou Yeh. Finally, to ensure a fair and standardized comparison, the evaluation phase in Section 3.5 directly employs the benchmark code provided by Search3D [1].

2. Related Works

Current approaches for segmenting instances from 3D scene representations typically fall into two main categories. The first category focuses on assigning visual features to whole-object predictions. These 3D object masks can be obtained via deep learning models, such as OpenMask3D [2] and PS3 [5], both leverages Mask3D [13] to generate 3D object-level predictions. However, the performance of these methods is heavily contingent on the quality of the initial object-level proposals. Furthermore, methods like OpenMask3D [2] often lack the flexibility to handle varying granularities, particularly for part-level instance segmentation.

The second approach involves aligning point-wise feature expressions with the underlying 3D representation. For instance, OpenGaussian [14] and InstanceGaussian [15] associate semantic feature vectors with each Gaussian point in a 3D Gaussian Splatting scene. While such point-wise representations theoretically support arbitrary granularity, existing 3DGS-based open-vocabulary methods can incur substantial memory overhead and suffer from noisy per-point features. Even with the instance-level aggregation strategy in InstanceGaussian [15], the resulting segments are not organized into an explicit, discrete instance hierarchy, which can make it difficult to obtain clean and consistent instance boundaries in complex scenes.

In the domain of 3D instance segmentation, a recent trend involves lifting 2D masks into 3D space by aggregating cross-view 2D masklets. These methods utilize 2D segmentation models, such as SAM [16] and SAM 2 [10], to isolate 2D instances, followed by feature extractors like CLIP [17] (and, in some recent works, SigLIP [11]) to obtain open-vocabulary visual features. Prominent examples include Open3DIS [4], which utilizes SAM [16], and Any3DIS [3], which leverages SAM 2 [10] to enable tracking capabilities across multiple frames. However, most 2D mask lifting approaches still focus primarily on object-level instances, making it non-trivial to support fine-grained, part-level reasoning in a fully training-free setting.

The advancement of these methods is underpinned by the evolution of transformer-based Segment Anything Models. Following SAM's [16] introduction of geometric prompting for class-agnostic masks, Semantic-SAM [9] introduced granularity control based on prompt specificity. On the other hand, SAM 2 [10] incorporates memory control and cross-frame attention mechanisms, breaking the single-image limitation and allowing users to track objects in video via key-frame prompting. In parallel, for image-text feature alignment, CLIP [17] revolutionized visual recognition, with successors like LAION-CLIP [12] and SigLIP [11] achieving superior performance through distinct dataset and loss function during training, respectively.

Finally, regarding open-vocabulary part-level instance segmentation, Search3D [1] proposes a hierarchical framework that has demonstrated effectiveness on datasets such as MultiScan [7] and ScanNet++ [6]. By over-segmenting object-level masks derived from Mask3D [13] and

constructing a tree structure connecting object and part levels, Search3D enables joint searches across both levels during text-based querying.

3. Method

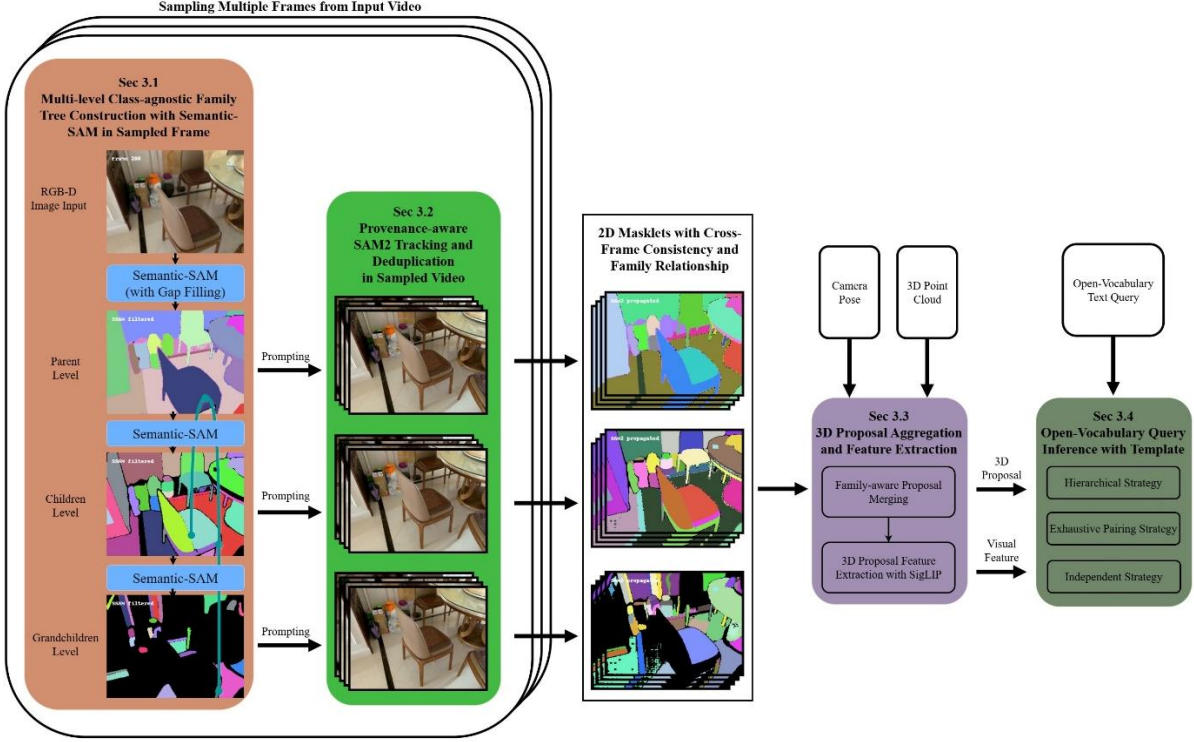


Fig. 1 Overview of the FamilyPart Pipeline.

We propose a new pipeline to segment 3D part-level instances with open-vocabulary querying, as shown in the figure above. In **Sec. 3.1**, we aim to obtain family relationships by progressively segmenting 2D RGB-D images. In **Sec. 3.2**, we retrieve inter-frame consistency, resulting in 2D masklets across the RGB-D image sequence. Then, on **Sec. 3.3**, we aggregate 2D masklets into 3D proposals, forming 3D binary masks over the point cloud of the scene and applying merging. Next, we extract open-vocabulary visual features of the proposals. Finally, we design three different algorithms to perform cross-level, part-level instance inference.

3.1 Multi-level Class-agnostic Family Tree Construction with Semantic-SAM

We leverage the hierarchical instance segmentation ability of Semantic-SAM. Semantic-SAM can generate class-agnostic instance masks for an image at six different levels, from coarse to fine, levels 1 to 6, respectively. However, the correspondence between the object and part levels in MultiScan and the granularity levels of Semantic-SAM is uncertain and therefore not fixed a priori. To address this, we construct a family-like relationship among coarse and fine masks.

Specifically, we select three levels to segment a single frame. For example, level 2 masks are regarded as parent nodes, level 4 masks as children nodes, and level 6 masks as grandchildren nodes. This family-based architecture ensures that we can retrieve parents with object features and find their offspring as part-level candidates during the inference stage.

3.1.1 Solo Photo Method



Fig. 2 A family of instance in a single frame. The images from left to right are original frame, one L2 instance, seven L4 instances, five L6 instances.

On the three levels of granularity (taking L2, L4, and L6 as an example), we build family trees in a single frame by progressively refining through the three levels. First, we prompt Semantic-SAM at level 2 to segment $\mathbf{K}$ instances, represented as $\mathbf{K}$ binary masks that share the same size as the image. Second, we conduct a logical **AND** operation between the original image and the masks, and pad black pixels as background, resulting in a “solo photo” of each level 2 instance. Next, we use the solo photos as input images and prompt Semantic-SAM at level 4 to segment their children. The output binary masks of the children are then logically **AND**ed with their parents, preventing Semantic-SAM from segmenting the black background and ensuring that the children masks are always inside their parents, maintaining the parent–child relationship. We also observe that some children have exactly the same appearance as their parents, which is redundant for hierarchical retrieval; therefore, we discard masks that are identical to their parents and those whose area is below a threshold. Finally, after applying the same procedure to obtain level 6 instances, we can construct the entire family tree and assign unique mask IDs that show family relationship via a lookup table.

3.1.2 Gap Filling



Fig. 3 Comparison before and after gap filling; the rightmost image is applied gap filling.

During our experiments, we observed that Semantic-SAM does not produce masks that completely fill the entire image. As a result, objects that are not segmented at the highest level may never be retrieved, since they are not included in any top-level mask. To mitigate this issue and avoid missing such objects, we additionally extract connected pixel regions that are not covered by any highest-level mask. When the area of such a region exceeds a predefined threshold, we record it as an independent mask. Note that we only apply this procedure at the highest level, because full-frame coverage is not necessary for the child levels. Some masks at finer levels may already be visually identical to their parents, in which case further subdivision is unnecessary; therefore, we do not enforce child masks to fill the entire image. If such a mask happens to contain multiple objects that Semantic-SAM failed to separate, they will be further partitioned at the next, finer level.

3.2 Provenance-aware SAM2 Tracking and Deduplication



Fig. 4 An example frame showing Semantic-SAM segments (middle) and the corresponding results after SAM2 tracking (right) in a single level.

In the previous stage, we constructed a Semantic-SAM-based family tree on each single frame using unique mask IDs. We now need to estimate the locations of these objects under different viewpoints and establish temporal consistency, which will serve as the basis for subsequent back-projection into 3D space. To maintain cross-frame consistency over the image sequence, we adopt the “propagate in video” functionality of SAM2, using the segment masks produced by Semantic-SAM in the previous stage as mask prompts. Starting from a single frame, SAM2 propagates these masks forward and backward in time, segmenting the corresponding objects in neighboring frames.

Since SAM2 does not allow overlapping masks within a single propagation pass, each granularity level must be tracked independently. In addition, the original frame rate provided by MultiScan is higher than required for our setting; naively tracking across all T frames would lead to excessive computational and memory costs. In practice, we apply a sampling strategy with a propagation stride of 50, i.e., we only retain one frame every 50 frames as a tracking node, resulting in approximately $(T/50)$ frames in total. Any3DIS [3] suggests that if an object

disappears for more than about seven frames before reappearing, the memory of SAM2 may fail. With the above sampling strategy, we reduce resource consumption while still maintaining a reasonable interval for object visibility.

3.2.1 Deduplicating and Virtual Children

Since we cannot guarantee that all target objects appear within a single frame, we perform progressive refinement over multiple frames. To allow Semantic-SAM to supervise the tracking results of SAM2, we adopt the deduplicate module proposed in Algorithm 1 of Semantic Consistent Language Gaussian Splatting for Point-Level Open-vocabulary Querying [8], whose core objective is to avoid repeatedly tracking semantically identical candidate objects in the SAM2 tracking stream.

Algorithm 1 Extracting regions with SAM and SAM2 [8]

```

1: Input: Image sequence  $I_{1:T}$ , segmentors SAM and SAM2, threshold  $\kappa$ 
2:  $\widetilde{\mathcal{S}}_{1:T} \leftarrow \{\}$  # Tracked masklets
3: for  $t \in \{1, \dots, T\}$  do
4:    $\mathcal{S}_{I_t} = \text{SAM}(I_t)$ 
5:   # Check if a region is tracked.
6:   for  $S_r \in \mathcal{S}_{I_t}$  do
7:     for  $\widetilde{S}_{r'} \in \widetilde{\mathcal{S}}_{1:T}$  do
8:       if  $\text{IoU}(\widetilde{S}_{r'}[t-1], S_r) > \kappa$  then
9:          $\mathcal{S}_{I_t} \leftarrow \mathcal{S}_{I_t} \setminus \{S_r\}$ 
10:      end if
11:    end for
12:  end for
13:  # Adding untracked masklets
14:   $\mathcal{B} \leftarrow \text{GetBoxFromMask}(\mathcal{S}_{I_t})$ 
15:   $\widetilde{\mathcal{S}}_{1:T} \leftarrow \widetilde{\mathcal{S}}_{1:T} \cup \text{SAM2}(I_{1:T}, \mathcal{B})$ 
16: end for
17: Output:  $\widetilde{\mathcal{S}}_{1:T}$ 

```

Concretely, we first use the Semantic-SAM masks segmented in the first frame as prompts for SAM2, and for each subsequent frame we compare the masks tracked by SAM2 with the Semantic-SAM masks of that frame using the Intersection over Union (IoU); if the IoU is high, we consider the object already tracked, whereas newly discovered objects are treated as prompts and passed to SAM2 for further propagation. Naïvely, this would require SAM2 to segment on the order of T^2 frames for an image sequence of length T . In practice, we reduce the computational burden by applying a subsampling strategy: we run Semantic-SAM at one fourth the frequency of SAM2, i.e., on roughly $T/200$ frames in a sequence of length T ; before

each batch of Semantic-SAM candidates is fed into SAM2, we extract the candidate masks, compare them against the already tracked masks using IoU and reject those with excessively high IoU, while recording the mask ID, source frame index, and packed mask representation so that subsequent frames can query whether a similar object has already been tracked in $\mathcal{O}(1)$ time and establish corresponding “virtual child” relationships.

The virtual-children mechanism consumes the rejection information produced by the deduplicate module and preserves it at the provenance level. For any rejected Semantic-SAM candidate, the system uses the mask ID returned by deduplicate module to uniquely determine which existing SAM2 object this candidate should be merged into and accordingly updates the lookup table. In this way, although the candidate is no longer propagated, its position in the family hierarchy is fully preserved in the form of a virtual child and can be traced back to its parent object in $\mathcal{O}(1)$ time through the same mapping. Regardless of whether a child object is rejected, the parent object ID it points to (including those of merged virtual children) will in turn point to the actual retained SAM2 object. This unified mapping design ensures that the semantic relationships across all levels of the family tree are not lost due to deduplicate, thereby achieving the dual goals of duplicate-free tracking and full lineage preservation.

3.2.2 Window-Based Method

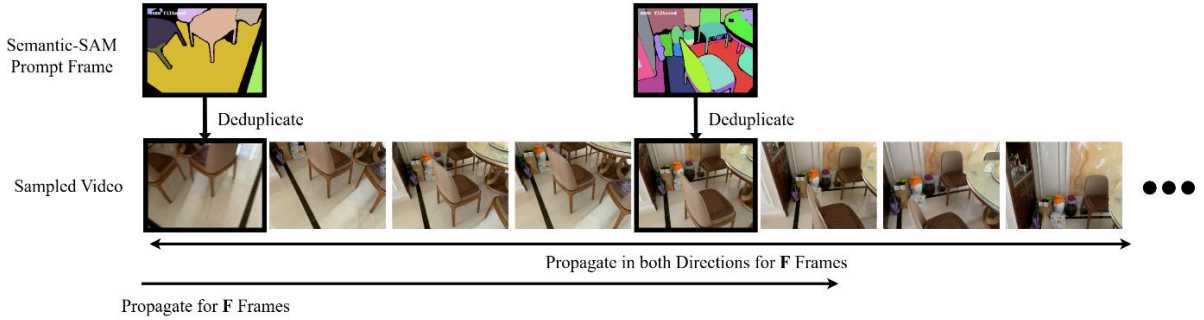


Fig. 5 The diagram of Window-Based Tracking in a single level.

Given the large number of frames in the dataset and the need to process three hierarchy levels, we impose an upper bound on the propagation length of SAM2 due to time and GPU memory constraints. For each selected Semantic-SAM prompt frame, we construct a subsequence of length $2F+1$ centered at that frame by extending F frames forward and F frames backward. Since SAM2 maintains an internal memory, the state initialization incorporates the feature cache of the entire subset video. Once this state is established, the memory of previously tracked objects is preserved even if the state is reset at the beginning of each new tracking pass. Thus, we run SAM2 propagation within each subsequence of length $2F+1$ and let the next prompt frame continue in the same manner, and we choose $F > 4$ (typically $F=30$), ensuring that the entire sequence is covered by multiple overlapping windows. Ideally, different frames observe the scene from varying viewpoints, and SAM2-based tracking can propagate

information across frames to achieve a complementary effect. In this way, the tracked object mask tends to approximate the union of its visible extents over all viewpoints. Therefore, there are more instances segmented after SAM2 stage in Fig. 4.

3.3 3D Proposal Aggregation and Feature Extraction

After obtaining the locations of each object and part across different frames, we then reconstruct them as 3D objects. We first query the family tree to retrieve all objects at the current hierarchy level, together with the list of frames in which each object appears; this serves as the basic unit for subsequent per-object projection. For each object, we take its 2D masklets across frames and, using the camera intrinsic and extrinsic matrices provided by the dataset, construct the corresponding transformation matrices. Combined with the depth information, these masks are then back projected into 3D space.

Next, we aggregate the per-frame 3D masks by taking their union in space to obtain 3D proposals, represented as binary arrays of length equal to the number of points in the scene point cloud. We then apply a minimum volume threshold to filter out small proposals. Finally, we perform family-aware merging on overlapping proposals using an IoU threshold of 0.5, which consolidates redundant proposals and yields a cleaner hierarchy of 3D proposals.

3.3.1 Family-aware Proposal Merging

The family-aware merging operates as follows. We first merge all top-level objects whose IoU exceeds a predefined threshold, and during each merge we reassign their child nodes so that they are attached to the same parent node. For lower-level nodes, merging is restricted to siblings within the same family tree: a child proposal is only compared with its sibling nodes for IoU-based merging, which prevents any child from being associated with multiple parents. After all merges are completed, we store the updated family relationships for use in subsequent stages of the pipeline.

3.3.2 3D Proposal Feature Extraction

After obtaining the 3D proposals, we perform open-vocabulary inference using transformer-based vision–language encoders such as LAION-CLIP (768 dimensions) and SigLIP (1152 dimensions). For each 3D proposal, we first query the occurrence table recorded in the SAM2 stage to obtain the list of frames in which the object appears and retain at most $\mathbf{M}$ frames. These frames are selected by uniform sampling over time so that the observations are not concentrated in a short interval with very similar viewpoints. In each selected frame, we compute the 2D bounding box of the object and crop around it at three different scales by expanding the box outward. This multi-scale cropping avoids overly tight crops that would truncate the object or include excessive background. The resulting image crops are then fed into the encoders to extract proposal features.

We adopt a feature pooling strategy to aggregate features across crops and frames. Specifically, we first apply L_2 normalization to each feature vector and then average them. This ensures that subsequent cosine similarity operates purely on the feature directions rather than their magnitudes. This pooling method is also used in OpenMask3D [2]. L_2 normalization can be depicted as follow:

$$\hat{\mathbf{f}} = \frac{\mathbf{f}}{\|\mathbf{f}\|_2} = \frac{\mathbf{f}}{\sqrt{\sum_{i=1}^D f_i^2}}, \quad \cos(\hat{\mathbf{f}}, \hat{\mathbf{t}}) = \frac{\hat{\mathbf{f}} \cdot \hat{\mathbf{t}}}{|\hat{\mathbf{f}}|_2 |\hat{\mathbf{t}}|_2}$$

For a feature vector $\mathbf{f} \in \mathbb{R}^D$, the L_2 -normalized vector $\hat{\mathbf{f}}$ is computed as shown above. Cosine similarity between a normalized proposal feature $\hat{\mathbf{f}}$ and a normalized text/query feature $\hat{\mathbf{t}}$ is then given by the second equation above. Because both vectors are normalized, the denominator equals 1, and retrieval can be implemented simply as a dot product. Across frames, we further weight the features according to a temporal Gaussian and the bounding-box area, leveraging both sequential and spatial information. Let $\mathbf{f}_k$ denote the feature from the k -th crop (or frame), and let w_k be its raw weight. We first compute normalized coefficients α_k by a softmax as follows.

$$\alpha_k = \frac{\exp(w_k)}{\sum_{j=1}^K \exp(w_j)}, \quad w_k = \exp\left(-\left(\frac{k-c}{\sigma}\right)^2\right) \sqrt{A_k}, \quad \mathbf{f} = \sum_{k=1}^K \alpha_k \mathbf{f}_k$$

Here, $c = (K - 1)/2$ is the temporal center of the frame sequence with length K , $\sigma = K/4$ controls the temporal spread of the Gaussian kernel, and A_k is the area of the k -th cropped bounding box. This aggregated feature vector is dominated by high-quality frames while smoothing temporal noise. Finally, we apply L_2 normalization once more so that the final proposal representation lies on the unit hypersphere.

3.4 Open-Vocabulary Query Inference with Template

In the MultiScan dataset, the open-vocabulary query text is defined as (object, part) tuples, such as (door, frame) or (cabinet, door). To obtain text features, we use the same vision–language model as in the 3D proposal pipeline, namely SigLIP or LAION-CLIP, as the text feature encoder. To enhance the semantic understanding of the model, we apply a textual template to the queries before tokenization: object-only queries are converted to “a {object} in a room”, while (object, part) pairs are converted to “a {part} of {object} in a room” (hereafter referred to as “combined queries”). For example, the pair (cabinet, door) becomes “a door of cabinet in a room”. Similar prompt engineering method can be seen in OpenMask3D [2] and PS3 [5]. The encoded text features are then L_2 -normalized and compared with the proposal features using cosine similarity, which reduces to an inner product under normalization. The

resulting similarity scores are scaled by a temperature parameter and passed through a SoftMax function to obtain a sharper and more discriminative distribution. The math relations are shown below. Here, q denotes the original text query (object or (object, part) tuple), and $T(q)$ is the textual template. The function $f_{\text{text}}(\cdot)$ is the text encoder, and $f_{\text{vis}}(\cdot)$ is the visual encoder applied to the i -th 3D proposal p_i . The vectors $\tilde{q}$ and $\tilde{p}_i$ are the L_2 -normalized text and proposal features, so the similarity score s_i is simply their cosine similarity. The scalar α is a temperature parameter that rescales the similarity scores before the softmax, and P_i is the final normalized matching probability of proposal p_i for the query q .

$$\tilde{q} = \frac{f_{\text{text}}(T(q))}{\|f_{\text{text}}(T(q))\|_2}, \quad \tilde{p}_i = \frac{f_{\text{vis}}(p_i)}{\|f_{\text{vis}}(p_i)\|_2}, \quad s_i = \tilde{p}_i^\top \tilde{q}, \quad P_i = \frac{\exp(\alpha s_i)}{\sum_j \exp(\alpha s_j)}$$

3.4.1 Hierarchical Strategy

Since the object and part levels are not explicitly aligned with any specific hierarchy level, we consider three levels (for example, L2, L4, L6) and design several querying strategies. The most straightforward strategy is a top-down, family-tree-guided search. We first retrieve the top- K_1 L2 candidates with the highest similarity to the object query. For each such L2 candidate, we then collect its children at L4 and add those proposals to the object candidate set if their similarity to the object query exceeds a threshold. Next, we compare the descendants (at L4 and L6) of these object candidates with the combined part-level query and add the top- K_2 proposals whose similarity exceeds a threshold to the part candidate set. After forming multiple (object, part) candidate pairs, we apply non-maximum suppression (NMS) but using only the part proposals as NMS masks (hereafter referred to as ‘‘Cross-path NMS’’). In this way, different object paths that point to the same physical part are treated as duplicates and suppressed, preventing multiple highly overlapping part predictions. This approach benefits from a reduced search space but is sensitive to the quality of the highest-level proposals: if the family tree is incomplete or incorrect, the search may fail.

3.4.2 Exhaustive Pairing Strategy

The second strategy exhaustively enumerates pairings within the family tree. Under the three-level hierarchy (L2, L4, L6), we consider six possible (object, part) level combinations: (L2, L2), (L2, L4), (L2, L6), (L4, L4), (L4, L6), and (L6, L6). For each combination, we retain only the top- K pairs with the highest similarity scores. We then validate these pairings in 3D space, discarding those where the object and part proposals do not overlap. The remaining candidates are passed to Cross-path NMS to obtain the final predictions.

3.4.3 Independent Strategy

The independent strategy decouples object and part retrieval from the family tree. We match the highest level (e.g., L2) against the object query, and the lower two levels (L4 and L6)

against the part query, each producing its own top-K proposal set. After removing cases where the same proposal is selected as both object and part, we evaluate the geometric relationship between each object–part pair by checking their overlap and relative size, and compute a geometric score based on IoU and size ratio. This geometric score is then combined with the original retrieval scores of the object and part queries via a weighted average. This is different from Search3D [1], which simply averages the cosine similarity of query text feature and 3D proposal feature on object and part level.

Although last two strategies have higher pairing complexity, they bypass the limitations of the family tree and remain effective even when the tree structure is noisy or incomplete.

3.5 Evaluation with Search3D Metric

We evaluate our method using the mean Average Precision (mAP) metric under the Search3D protocol. For each IoU threshold τ , we rank all predictions by confidence score, compute the corresponding precision–recall curve, and integrate it to obtain $AP(\tau)$. The metrics **AP25** and **AP50** correspond to AP at $\tau = 0.25$ and $\tau = 0.50$, respectively. The metric **AP** is the mean of AP over multiple IoU thresholds from 0.50 to 0.95 with a step size of 0.05. Under the Search3D protocol, $mAP(\tau)$ is computed per part-level label and then macro-averaged across categories. Therefore, **mAP** reflects the overall mean precision across all categories at the part level.

4. Experiments

4.1 Implementation Details

We run our multi-scene pipeline on the MultiScan dataset using the open-vocabulary 3D part benchmark from Search3D, which expands the original 17-object/5-part labels to 155 object and 15 part categories plus 47 (object, part) pairs and process all scenes in each official split. We use the LAION-CLIP ViT-L/14 checkpoint (laion/CLIP-ViT-L-14-laion2B-s32B-b82K) with score scale 300 and softmax scoring, and SigLIP (google/siglip-so400m-patch14-384) with scale 200. Aggregation operates on three hierarchy levels and discards 3D proposals whose visible area fraction is below 0.0005. The Semantic-SAM stage samples every 4th frame, applies Semantic-SAM with the swinl_only_sam_many2many checkpoint, uses area threshold=100; tracking uses SAM2 (sam2.1_hiera_large) with IoU threshold=0.6, downscale ratio=0.3. All experiments export per-scene metrics and run on CUDA GPUs.

4.2 Results and Analysis

Table. 1 AP Results With Different Family Levels (laion-clip, min. area =500) Unit: %

Level	Independent mAP / AP50 / AP25	Hierarchical mAP / AP50 / AP25	Exhaustive mAP / AP50 / AP25
L1/L3/L5	1.2 / 3.7 / 8.0	0.0 / 0.1 / 4.3	1.1 / 3.2 / 11.0
L2/L4/L6	0.3 / 1.6 / 6.8	0.9 / 2.3 / 10.7	0.0 / 0.2 / 7.8

(Reference: Experiment)

Table. 2 AP Results With Different Feature Extractor (L1/3/5, min. area=100) Unit: %

Model	Independent mAP / AP50 / AP25	Hierarchical mAP / AP50 / AP25	Exhaustive mAP / AP50 / AP25
laion CLIP-ViT-L-14	1.2 / 3.7 / 11.4	0.0 / 0.1 / 3.8	0.5 / 1.4 / 8.3
siglip-so400m	1.1 / 3.1 / 11.8	0.1 / 0.4 / 6.6	0.9 / 2.7 / 9.7

(Reference: Experiment)

Table. 3 AP Results With Different Inference Strategies over 7 Experiments Conducted
Unit: %

Strategies	mAP	AP50	AP25
independent	0.84	2.90	8.59
hierarchical	0.25	0.75	7.01
exhaustive	0.51	1.54	8.39

(Reference: Experiment)

From the experimental results, we observe that among the 41 scenes, fewer than half achieve a non-zero mAP (excluding four scenes without ground-truth annotations). The hierarchical strategy performs particularly poorly: only one scene attains a non-zero mAP, and the performance variance across scenes is large. This sensitivity is closely related to the number of video frames per scene and to the inherent size and spatial distribution of objects and parts. However, since each scene contains only a limited number of annotated instances, even a small number of high-quality predictions can already yield a relatively high AP, making the absolute prediction count less critical. Overall, our best configuration (L135, min. area = 100, SigLIP, Independent strategy) yields mAP/AP50/AP25 scores of 1.0/3.4/14.7. This is still notably behind Search3D [1] (7.9/14.5/31.5).

The primary issue lies in the deduplication logic used in the SAM2 stage. In our implementation, we rely solely on IoU as the criterion for discarding masks. This is appropriate in the original framework of [8], which operates at the object level, where relative size can serve as a reasonable heuristic for suppression. However, when dealing with different

granularities, using only IoU leads to the loss of size relationships. For example, even when Semantic-SAM is prompted at the same hierarchy level, the degree of refinement for the same object can vary across viewpoints, so after propagation the same object may be represented by masks of different granularity across frames. When these masks are compared and deduplicated purely by IoU, masks at different scales are likely to suppress each other, effectively destroying the intended hierarchical structure. As a result, the assumption that children are always contained inside their parents no longer holds, and we occasionally observe part masks that are larger than their corresponding object masks. This, in turn, causes the hierarchical method to fail almost completely. In addition, due to hardware constraints, we increased the minimum area threshold to save memory; this leads to many fine-grained masks being discarded and further degrades part-level performance. Consequently, the more fine-grained L2/L4/L6 combination performed worse, as it was disproportionately affected by this aggressive filtering.

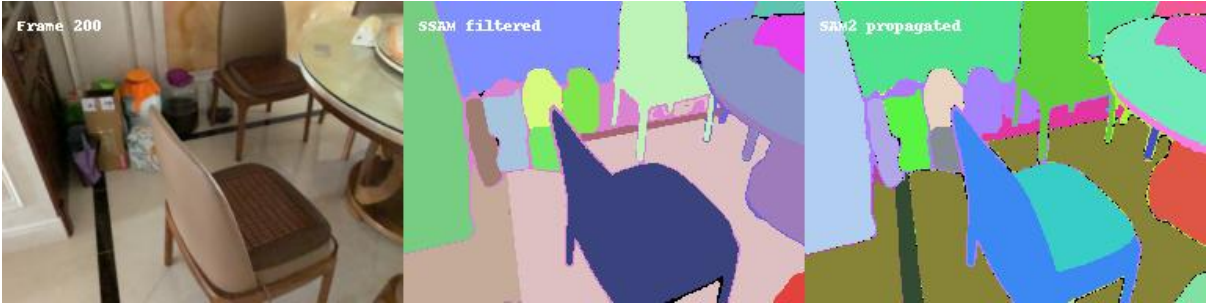


Fig. 6 An example of change in granularity after SAM2 tracking (at the chair).

During implementation, we also observed that SigLIP and LAION-CLIP alone may be insufficient to achieve tight feature alignment between 3D proposals and text queries. Even when using ground-truth 3D masks for matching against text queries, not every label attains a similarity score above 0.8, and there is considerable variation across labels.

Regarding the matching strategies and models, the independent strategy achieves the best AP. Given the poor quality of the extracted family trees, both the independent and the exhaustive pairing strategies are less affected than the hierarchical one. However, because the number of prediction pairs and their confidence scores directly influence mAP, exhaustive pairing tends to generate an excessive number of pairs, which dilutes the confidence mass over many less informative pairs, such as those between instances at the same level. In contrast, the independent strategy can focus more robustly on discovering and ranking pairs with stronger semantically relevant pairs. As for the encoders, SigLIP generally outperforms the CLIP-based LAION-CLIP, consistent with its original claims. LAION-CLIP only surpasses SigLIP in mAP and AP50 under the independent strategy. We hypothesize that this may indicate that LAION-CLIP is better at classifying certain local object details, likely due to being trained on an even larger dataset, and therefore can more effectively identify highly overlapping 3D proposals for

some object categories.

4.3 Future Directions

Due to time and computational constraints, we were unable to conduct more extensive comparative experiments or ablation studies. Nevertheless, based on our current findings, we identify several promising directions for future improvement. First, to address the size-agnostic deduplication issue, a new module is needed to distinguish whether two masks stand in a containment relationship or are truly identical. This module should explicitly account for relative mask size when deciding whether a candidate should be preserved as a child, rather than relying solely on IoU. Second, a scale-adaptive filtering scheme should be adopted, so that different granularity levels do not share a single, global minimum-area threshold, which currently causes many fine-grained part-level masks to be discarded. Finally, our current scoring mechanism is relatively simple; incorporating more sophisticated weighting or feature fusion strategies may further improve mAP in future work.

5. Conclusion

In this work, we propose a family-tree-based hierarchical open-vocabulary 3D part-level instance segmentation framework. We first perform multi-level progressive semantic segmentation with Semantic-SAM to construct parent-child relationships between nested masks. To obtain cross-frame consistency under varying viewpoints, we then use the Semantic-SAM masks as prompts to supervise SAM2’s propagate in video tracking. The resulting 2D masklets are aggregated into 3D proposals, on which we apply open-vocabulary feature extraction using SigLIP and related vision-language models. Finally, we design several inference strategies to perform retrieval over the 3D proposals.

Unfortunately, design flaws in the 2D mask propagation stage lead to degraded AP scores. Nonetheless, our study demonstrates the flexibility introduced by multi-level family relationships and retrieval strategies and shows that leveraging existing foundation models for this task is highly promising. We believe that further improvements to the deduplication module, scale-adaptive filtering, and score fusion mechanisms could substantially enhance both the accuracy and practical usability of the proposed framework.

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7. Project Management and Teamwork

7.1 Plan Management

This research topic grew out of my personal curiosity about 3D instance segmentation (3DIS). After joining the Vision Science Lab, I first surveyed the literature on 3DIS and participated in the research projects of master’s students. The method and pipeline in this thesis were gradually shaped through brainstorming with my mentors in the lab. Later, PS3 was developed by graduate student Yen Hong-Xuan, and I collaborated with graduate student Ou Yeh on this project as my implementation project.

During the implementation stage, we held weekly online meetings to report progress and address practical issues, such as memory management and feature post-processing tips. The overall research framework and experimental design of this work were individually planned and carried out by myself, while parts of the code implementation were adapted from other sources. I would like to thank my mentors in the lab, including Yen Hong-Xuan, Ou Yeh, Chen Chia-Min, and Wang Yan-Qing, for their valuable discussions and guidance.s

7.2 Teamwork

No other undergraduate students participated in this research project. The detailed division of labor and the originality of this work are described in the Introduction section and are therefore not repeated here.